Targeted SAT Math — Set 07

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Homework after Suite Hard Module 03. Five questions. Work on paper. Calculator / Desmos is fine.

These are **new** questions on two leftover jobs — not those official questions, and not copies from Sets 01–06.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

Write the job first.

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1. Notes for this set only

Vertical line vs horizontal line

A line whose equation is $x = k$ (for example $x = -5$) is **vertical**. Its slope is **undefined**. It is parallel to the y-axis.

A line whose equation is $y = k$ is **horizontal**. Its slope is **0**. It is parallel to the x-axis.

Perpendicular to a vertical line is horizontal, so that slope is **0**.
Perpendicular to a horizontal line is vertical, so that slope is **undefined**.

Undefined belongs to the **vertical line itself**. Do not give "undefined" to the line that is perpendicular to it.

Margin of error

A poll result like $45\% \pm 4\%$ means the true percent in the **whole population** is likely between 41% and 49%. It does **not** mean "exactly 45%."

SAT will not ask you to compute a formula. They ask you to read the interval, or to say what makes the margin of error bigger or smaller.

- Larger random sample $\Rightarrow$ **smaller** margin of error.
- Smaller random sample $\Rightarrow$ **larger** margin of error.

If two random samples come from the **same** population and sample A has the larger margin of error, sample A was the **smaller** sample. Not because it had a higher percent "in favor." Not because it was the larger sample.

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2. Questions

1. Line ℓ in the xy-plane is perpendicular to the line with equation $x = -5$. What is the slope of line ℓ?

A) -5
B) $-\frac{1}{5}$
C) 0
D) The slope of line ℓ is undefined.

2. What is the slope of the line with equation $x = 9$?

A) 0
B) $\frac{1}{9}$
C) 9
D) The slope is undefined.

3. Two random samples of votes on the same city measure were selected from the same population of voters. The margins of error were calculated the same way.

Sample	**Percent in favor**	**Margin of error**
A	54%	4.8%
B	50%	1.5%

Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

A) Sample A had a higher percent of favorable responses.
B) Sample A had a larger sample size.

C) Sample A had a smaller sample size.
D) Sample A had fewer ballots that could not be recorded.

4. A random sample of residents in a town shows that 38% support a new park, with a margin of error of 3%. Which conclusion is most appropriate?

A) Exactly 38% of all residents in the town support the park.
B) The true percent of all residents who support the park is likely between 35% and 41%.
C) The park will be approved.
D) The sample is biased because the result is not 50% to 50%.

5. Two random samples are taken from the same city. Sample P has 80 people. Sample Q has 800 people. The same method is used to compute each margin of error. Which sample is likely to have the **smaller** margin of error?

A) Sample P
B) Sample Q
C) The two margins of error must be equal.
D) It cannot be determined without the sample percents.